

## **APPENDICES: TASK 2**

### **APPENDIX 1: Calculation of Mobility and Equity**

#### ***Mobility and Equity Results on Route 169***

##### **Entrance Ramps**

In the network of Highway 169 north bound from I-494 to I-694, there are 15 metered entrance ramps which have sufficient data. The mobility measures used for entrance ramps are total delay and average delay for each individual vehicle. The results presented here are for ramps identified as 3A1, 3A4, 3B2, 3B3, 3B4, 3B5, 3B6, 3C6, 3C7. These ramps had visual observation of queue lengths recorded. The data are for March 21, 2000.

To illustrate the method, Figure A1-1 shows that for Ramp 3A1, the delay increases steadily till it reaches 68 seconds for the 48th car, again decreasing afterwards. It increases again to reach a peak delay of 203 seconds for the 773rd. In the process the delay follows both intermittently increasing and decreasing trends. Thereafter, the delay starts decreasing... which again is not a uniform process till it reaches 0 for the 1456th car. As we can see, there is significant variation in the delay values throughout the metering time.

The equity implications of the above analysis can be seen by viewing Table A1-1. This table shows the total delay, number of delayed vehicles, average delay, and redundancy and Gini coefficient equity statistics. The Gini coefficient (1 is perfect inequity, 0 is perfect equity) vary from 0.27 to 0.59 for the 9 ramps analyzed so far. The average delay for the ramps varies from 95 to 230 seconds. There is no direct correlation between the Gini coefficient and average delay on ramp segments.

##### **Freeway Segments**

This section shows show the changes of mobility and equity for the freeway segments between the ramp on case and the ramp off case. We can see that the average travel speed with ramp metering (Ramp On) is 61.5 mph while without (Ramp Off) it is only 26.1 mph. The average travel delay per mile in the ramp on case is only 8.9 sec/mile, but in ramp off case, it is as long as 126.4 sec/mile. So, it's obvious that the mobility of freeway segments increases significantly because of ramp control. (We also found that the speed differences between the two lanes range from 7.2 mph to 18 mph.)

The following two figures will help us to see the impact of ramp control on freeway segment equity. Figure A1-2 shows the relationship between the mobility measure (average travel speed) and the temporal equity measure (Gini coefficient).

Recall, the lower the Gini coefficient, more equity there is, that is the more equally spread is the delay.

The average travel speeds for all the 14 freeway segments in the Ramp On case are much higher than those in the Ramp Off case. In contrast with equity measures for OD trips (shown below), the Gini coefficients in the Ramp On case are smaller than those in Ramp Off case which shows that the Ramp On case is more equitable. So, the conclusion is that the relationship between mobility and temporal equity in freeway segments is positive.

Figure A1-3 illustrates the relationship between the mobility measure (average travel speed) and the spatial equity measure (Gini coefficient). The trend of mobility and equity is the same as that in Figure A1-2. We again see a positive relationship between mobility and spatial equity.<sup>1</sup> Table A1-2 shows the freeflow travel times and speeds for each freeway segment under the two cases, with and without ramp metering.

### ***Mobility and Equity Results on I-94***

By the same methodology as the Route 169 analysis, mobility and equity calculation results for I-94 are shown. The chosen 9.1mile segment is from TH55 to Marion St. which consists of nine entrance ramps and eight freeway segments.

#### **Free Flow Speed**

Again, freeway delay is defined as the difference between free flow travel time and actual travel time. The free flow travel time of a specified freeway segment is defined as the shortest travel time in this segment of all the five minute intervals. Table A1-3 gives the free flow data we obtained (based on an assumption of 26ft vehicle effective length.\*). The free flow speed here is higher than that on Route 169.

It seems that the free flow speeds of segment 2, 3, 5 and 6 are too high. A possible reason would be that the actual effective vehicle lengths in these segments are shorter than 26ft.

#### **Freeway Segment Mobility and Equity**

This section shows the changes of mobility and equity for the freeway segments between the ramp on case and the ramp off case. In contrast with the OD trip results, we can see the improvement of mobility in the ramp metering-on case. The average travel speed with

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<sup>1</sup> Note that if the mobility measure used in the above two figures is delay per mile not average travel speed, the relationship between mobility and equity is hard to tell. The probable reason is that the difference of means of delay per mile in the two ramp metering cases (on & off) is too large. The average delay per mile in the Ramp On case is only 8.9 sec/mile while in the Ramp Off case it's 126.4 sec/mile. We know that two things contribute to the change of Gini coefficient: mean and variance. So, when we use Gini coefficient as an equity measure, the smaller the difference between the means of two series of data, the more reliable of the results.

\* If using 22ft, the free flow speed will decrease proportionally.

ramp metering on case is 61.4 mph while it is only 53.9 mph with ramp metering shut off. The average travel delay per mile in ramp on case is only 12.1 sec/mile, but in metering off case, it is as long as 29.9 sec/mile. So, it's obvious that the mobility of freeway segments increases significantly because of ramp control.

Figure A1-4 shows the relationship between the mobility measure (average travel speed) and the temporal equity measure (Gini coefficient). The average travel speeds for 7 of the 8 freeway segments (except segment 6) in ramp on case are higher than those in ramp metering off case. The relationship between mobility (speed) and equity (Gini coefficient) is not very clear in the figure. Figure A1-5 illustrates the relationship between the mobility measure (average travel speed) and the spatial equity measure (Gini coefficient). The equity measure has a positive relationship with speed.

**Table A1-1: Mobility and Equity Measure on Entrance Ramps on Route 169**

| Entrance Ramp on 169 |                  | Total Delay(h) | # of Delayed Veh | Average Delay(sec) | Redundancy | Gini Coeficient |
|----------------------|------------------|----------------|------------------|--------------------|------------|-----------------|
| 3A1                  | Valley View Rd   | 43.58          | 1464             | 107.16             | 0.018382   | 0.275729        |
| 3A4                  | Bren Rd          | 72.75          | 1138             | 230.14             | 0.021911   | 0.300235        |
| 3B2                  | Excelsior Blvd   | 81.33          | 1407             | 208.09             | 0.013703   | 0.273634        |
| 3B3                  | T.H.7            | 37.36          | 1410             | 95.387             | 0.03852    | 0.412067        |
| 3B4                  | 36th St          | 37.50          | 715              | 188.81             | 0.09365    | 0.594324        |
| 3B5                  | Minnetonka Blvd  | 27.32          | 566              | 173.77             | 0.050115   | 0.441118        |
| 3B6                  | Cedar Lake Rd    | 23.00          | 455              | 181.98             | 0.083469   | 0.550404        |
| 3C6                  | Plymouth Ave     | 31.08          | 773              | 144.75             | 0.057886   | 0.440563        |
| 3C7                  | Medicine Lake Rd | 46.21          | 962              | 172.93             | 0.036535   | 0.370871        |

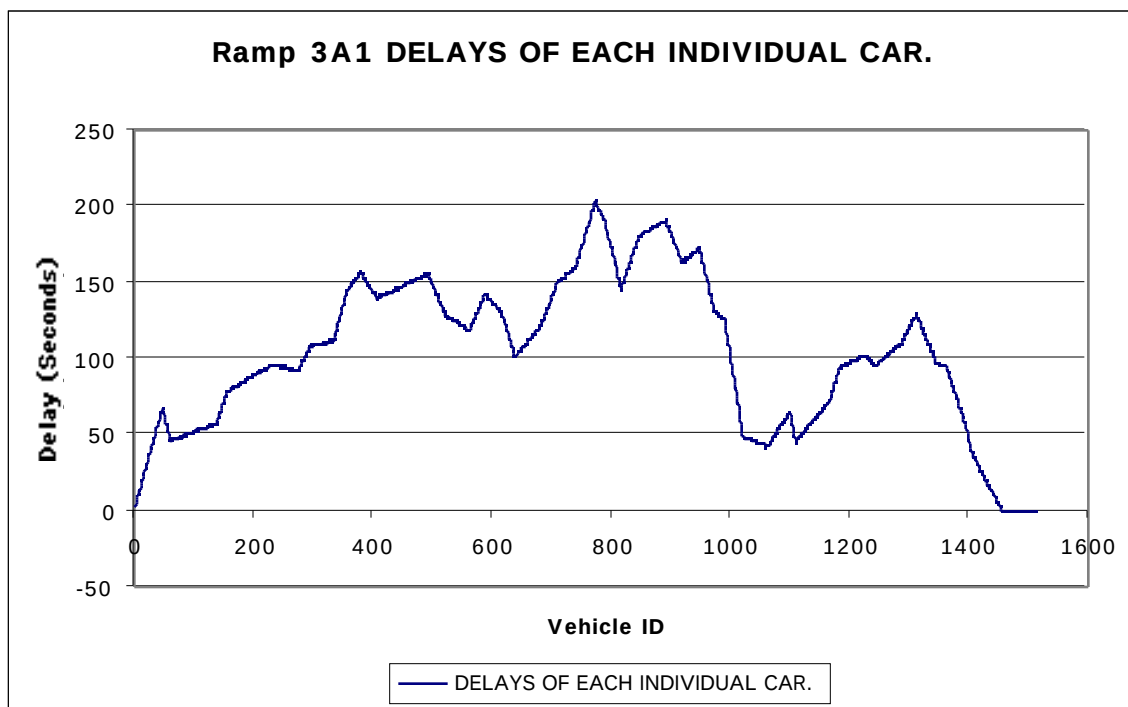
|                       |                 |
|-----------------------|-----------------|
| Redundancy of 9 sites | Gini of 9 sites |
| 0.015034              | 0.124947        |

**Table A1-2: Free Flow Travel Time and Speed: Ramp On and Ramp Off Cases, Route 169**

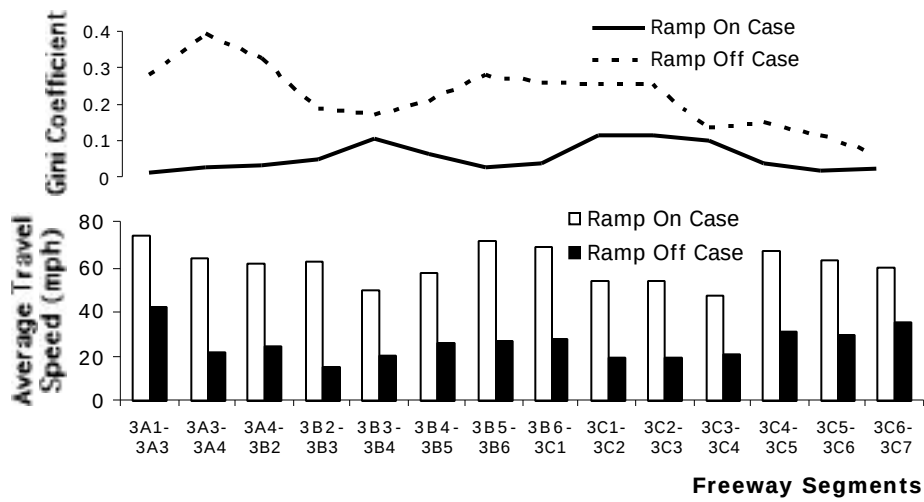
| From | To  | Segment length (Mile) | Ramp off free flow Travel Time(Sec) | Ramp on free flow travel time (sec) | Ramp off free flow speed (mph) | Ramp on free flow speed (mph) |
|------|-----|-----------------------|-------------------------------------|-------------------------------------|--------------------------------|-------------------------------|
| 3A1  | 3A2 | 1.42                  | 71.7                                | 64.8                                | 71.5                           | 79.2                          |
| 3A3  | 3A4 | 0.55                  | 33                                  | 28.6                                | 60.2                           | 69.5                          |
| 3B2  | 3B3 | 0.74                  | 77.5                                | 37.9                                | 34.3                           | 70.1                          |
| 3B3  | 3B4 | 0.3                   | 32.9                                | 18                                  | 32.5                           | 59.5                          |
| 3B4  | 3B5 | 0.43                  | 35.4                                | 23.3                                | 43.7                           | 66.4                          |
| 3B5  | 3B6 | 0.39                  | 25.7                                | 18.3                                | 55.2                           | 77.4                          |
| 3B6  | 3C1 | 1.33                  | 81.9                                | 63.7                                | 58.3                           | 75                            |
| 3C1  | 3C2 | 0.22                  | 15.7                                | 12.1                                | 51.3                           | 66.6                          |
| 3C2  | 3C3 | 0.19                  | 13.2                                | 10.1                                | 51.3                           | 67                            |
| 3C3  | 3C4 | 0.26                  | 22.1                                | 16.3                                | 42.9                           | 58.3                          |
| 3C4  | 3C5 | 0.24                  | 13.9                                | 11.2                                | 61.6                           | 76.2                          |
| 3C5  | 3C6 | 0.6                   | 48.6                                | 32.3                                | 44.4                           | 66.8                          |

**Table A1-3: Free Flow Travel Time and Speed, I-94**

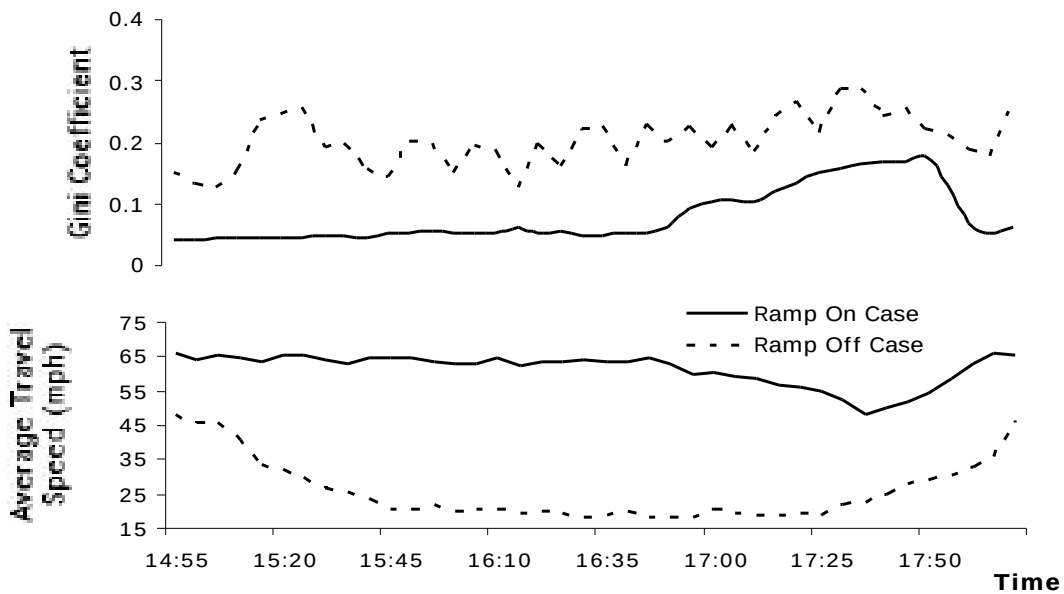
| Freeway Segment       | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
|-----------------------|------|------|------|------|------|------|------|------|
| Segment length (mile) | 1.34 | 0.87 | 0.74 | 0.69 | 1.81 | 2.04 | 0.87 | 0.72 |
| Freeflow TT (sec)     | 77   | 34.8 | 31.9 | 36.4 | 80   | 80.3 | 49.6 | 42   |
| Freeflow Speed (mph)  | 62.8 | 90.2 | 83.3 | 68.6 | 81.5 | 91.3 | 63.3 | 61.4 |



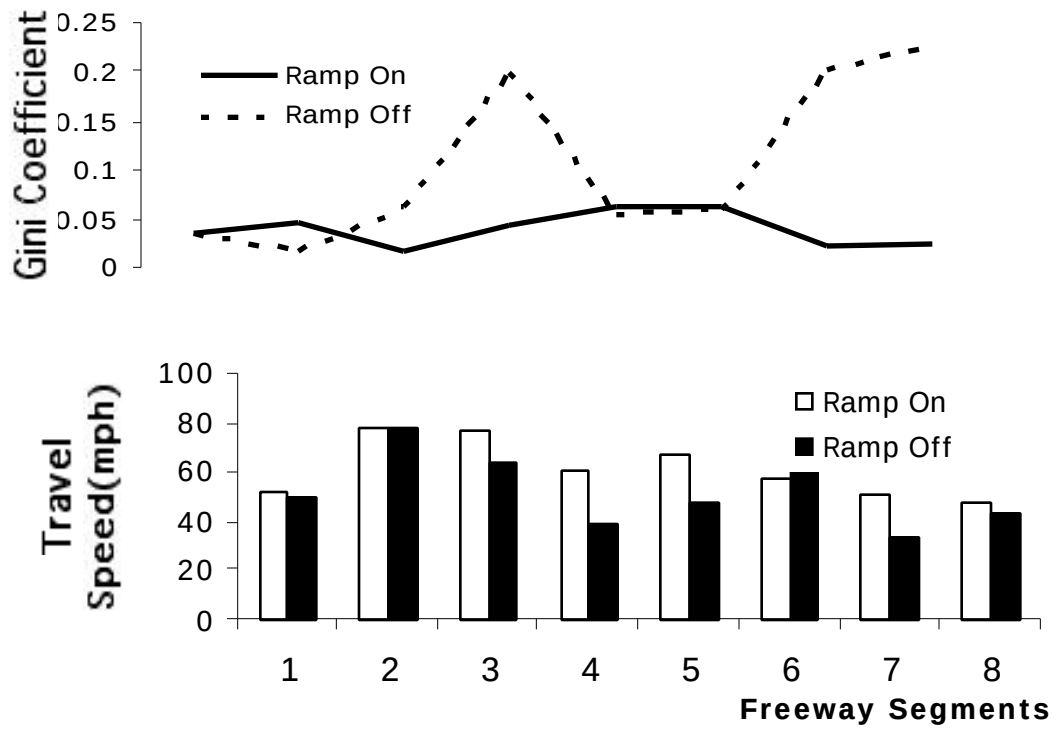
**Figure A1-1: Delays on Ramp 3A1, Route 169**



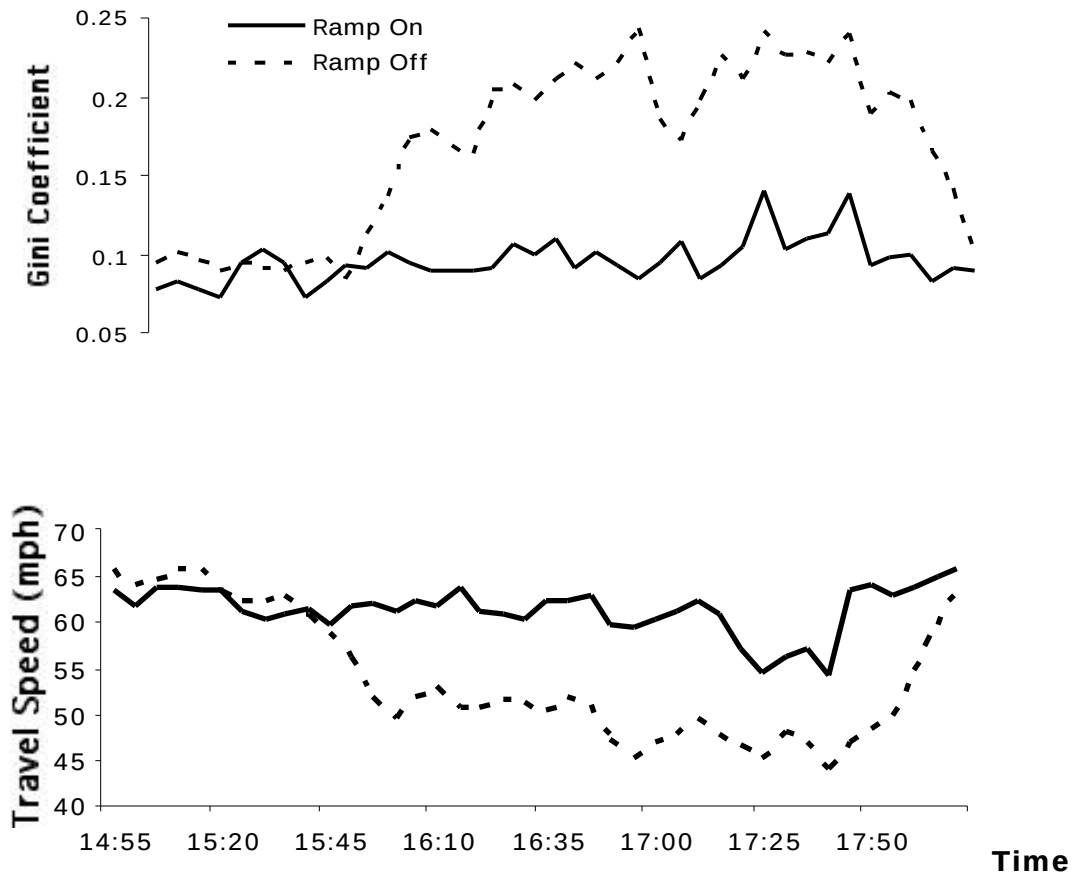
**Figure A1-2: Relationship Between Average Travel Speed and Temporal Gini Coefficient on Freeway Segments on Route 169**



**Figure A1-3: Relationship Between Average Travel Speed and Spatial Gini Coefficient on Freeway Segments on Route 169**



**Figure A1-4. Average Travel Speed and Temporal Gini coefficient, I-94**



**Equity**  
**Figure A1-5 Average Travel Speed and Spatial Gini Coefficient, I-94**



## APPENDIX 2: Calculation of Consumer's Surplus and Productivity

For each of the freeway segments the 5-minute flows and the travel times required for each vehicle was considered and the corresponding VKTs and VSTs were estimated. The VKTs and the VSTs for all the 5-minute intervals were added up to get the totals for each of the freeway segments and the ratio between the total VKT and VST gave the productivity for that freeway segment. The step was done for both the metered and the non-metered cases. The changes in flow and travel times on the freeway segments between the metered and the non-metered cases for every corresponding 5 minute let us calculate the change in consumer surplus for each 5-minute interval for any freeway segment (as explained above). Addition of all the changes (over all the 5 minute periods) gave the total change in consumer surplus for that freeway segment for the entire metered period.

Table A2-1 is a sample of calculations for one ramp (Olson Highway). Referring to Table A2-1, the first time interval starts at 14:55 and the last one is at 18:10. As can be seen, the first column is the ramp flows, the second one is ramp delay i.e., the number of seconds each car waits on the ramp. The third column is the total delay of the vehicle including ramp wait. The fifth column is the VKT (the product of the ramp flows with the length of the ramp in km, here it is 0.468 km). The last column is the VST (the product of the flows and the total delay on the ramp).

There is some need to explain the calculation in the third column. The values in this column include the stopped delay that each car has to experience on the ramp while being in queue plus the time that it takes in order to reach its position in the queue from the start of the ramp. This was calculated as follows:

The rates at which the cars were released from the ramps onto the highway depend on the red times. For each of the ramps, the red times were averaged across the whole metering period. For the example shown in Table A2-1, (Olson Highway), the red times for the entire period of metering was averaged and found to be 3.19 seconds. So each car is released every 3.19 seconds. If, for the first 5-minute interval, the ramp wait was 8.2 seconds, then there must have been  $(8.2/3.19 \Rightarrow) 2.57$  cars ahead of each car arriving in that 5-min interval (assuming that one car is allowed to leave every 3.19 seconds). Inter-car distance has been assumed as 6.1m and so the distance that each car has to traverse while being in queue is equal to  $2.57 \times 6.1 = 15.68$  m and the delay while covering this distance is already covered in the ramp wait. So the car has to cover the remaining distance (excluding the queue length) at a speed of 40 km/h (assumed). For bringing in some temporal uniformity, we have divided the total ramp wait (in queue) by the total number of 5-minute time intervals i.e., 40 and the result by 3.19 (for one car is released every 3.19 seconds) to get the average number of cars in the queue. In the sample shown in table 2, the average number of cars at any point of time in queue is 7.1 (each of which is released every 3.19 seconds) and so the rest of the distance is equal to the length of the ramp minus the length of 7.1 cars and the time required to cover that distance is equal to  $\{468 - (7.1 \times 6.1)\} / 11.11 = 38.2$  secs. [the cars travel the distance at 40 km/h=11.11m/s]. So the total time that a single car waits (in the first 5 minute interval) is

equal to  $(8.2+38.2=)$  46.43 seconds. This way all the total delays on the ramp are calculated. The delays multiplied by the flows give the VSTs and the flows multiplied by the length of the ramp give the VKTs. We sum up all the VKTs and VSTs to get the value for the entire period of metering and then the ratio VKT/VST gives the productivity. In the example shown above, the productivity of the ramp is equal to 0.0076. Likewise we find the productivities of all the ramps. However, for some ramps, the flows might spill over the ramps (when the total length of the vehicles in queue exceeds the ramp length) and then the total delay would simply be the ramp waits while in the queue. This has been incorporated in another column titled, "Average travel time including ramp waits (corrected)". For the non metered case, the total delay is simply the time required to travel the ramp length at 40 km/hour. And the rest of the calculations are identical to the method described above.

Some additional summary tables are provided:

- Table A2-2 : Analysis of freeway segments on US 169
- Table A2-3: Analysis of freeway segments on I 94
- Table A2-4: Ramp Productivity on US 169.
- Table A2-5: Ramp Productivity on I-94

## Appendix :

**Table A2-1: Olson Highway ramp analyzed:**

**Olson Hwy (Detector : 724) Length = 0.468 km**

| Time  | Ramp flows | Average waits on ramp (Seconds) | Average travel time including ramp waits (secs) | VKT (=Ramp flow *Length) | VST (= Flow * Avg Travel time) |
|-------|------------|---------------------------------|-------------------------------------------------|--------------------------|--------------------------------|
| 14:55 | 60         | 8.20                            | 46.43                                           | 28.1                     | 2786                           |
| 15:00 | 69         | 10.40                           | 48.63                                           | 32.3                     | 3355                           |
| 15:05 | 68         | 10.80                           | 49.03                                           | 31.8                     | 3334                           |
| 15:10 | 82         | 10.50                           | 48.73                                           | 38.4                     | 3996                           |
| 15:15 | 81         | 8.80                            | 47.03                                           | 37.9                     | 3809                           |
| 15:20 | 89         | 9.20                            | 47.43                                           | 41.7                     | 4221                           |
| 15:25 | 77         | 8.40                            | 46.63                                           | 36.0                     | 3590                           |
| 15:30 | 94         | 7.60                            | 45.83                                           | 44.0                     | 4308                           |
| 15:35 | 97         | 5.20                            | 43.43                                           | 45.4                     | 4212                           |
| 15:40 | 82         | 12.10                           | 50.33                                           | 38.4                     | 4127                           |
| 15:45 | 106        | 13.90                           | 52.13                                           | 49.6                     | 5525                           |
| 15:50 | 118        | 16.90                           | 55.13                                           | 55.2                     | 6505                           |
| 15:55 | 103        | 13.40                           | 51.63                                           | 48.2                     | 5318                           |
| 16:00 | 105        | 13.80                           | 52.03                                           | 49.1                     | 5463                           |
| 16:05 | 84         | 11.20                           | 49.43                                           | 39.3                     | 4152                           |
| 16:10 | 88         | 18.20                           | 56.43                                           | 41.2                     | 4966                           |
| 16:15 | 113        | 14.80                           | 53.03                                           | 52.9                     | 5992                           |
| 16:20 | 94         | 13.10                           | 51.33                                           | 44.0                     | 4825                           |
| 16:25 | 106        | 9.00                            | 47.23                                           | 49.6                     | 5006                           |
| 16:30 | 90         | 7.40                            | 45.63                                           | 42.1                     | 4106                           |
| 16:35 | 80         | 12.90                           | 51.13                                           | 37.4                     | 4090                           |
| 16:40 | 105        | 10.60                           | 48.83                                           | 49.1                     | 5127                           |
| 16:45 | 118        | 12.80                           | 51.03                                           | 55.2                     | 6021                           |
| 16:50 | 107        | 21.60                           | 59.83                                           | 50.1                     | 6401                           |
| 16:55 | 119        | 30.70                           | 68.93                                           | 55.7                     | 8202                           |
| 17:00 | 114        | 23.60                           | 61.83                                           | 53.4                     | 7048                           |
| 17:05 | 108        | 22.60                           | 60.83                                           | 50.5                     | 6569                           |
| 17:10 | 122        | 32.00                           | 70.23                                           | 57.1                     | 8568                           |
| 17:15 | 109        | 15.70                           | 53.93                                           | 51.0                     | 5878                           |
| 17:20 | 119        | 11.00                           | 49.23                                           | 55.7                     | 5858                           |
| 17:25 | 114        | 61.20                           | 99.43                                           | 53.4                     | 11335                          |
| 17:30 | 98         | 124.70                          | 162.93                                          | 45.9                     | 15967                          |
| 17:35 | 95         | 123.80                          | 162.03                                          | 44.5                     | 15393                          |
| 17:40 | 93         | 97.10                           | 135.33                                          | 43.5                     | 12585                          |
| 17:45 | 118        | 38.80                           | 77.03                                           | 55.2                     | 9089                           |
| 17:50 | 105        | 10.40                           | 48.63                                           | 49.1                     | 5106                           |
| 17:55 | 98         | 9.70                            | 47.93                                           | 45.9                     | 4697                           |
| 18:00 | 81         | 8.60                            | 46.83                                           | 37.9                     | 3793                           |
| 18:05 | 77         | 7.80                            | 46.03                                           | 36.0                     | 3544                           |
| 18:10 | 87         | 7.80                            | 46.03                                           | 40.7                     | 4004                           |
|       | 3873.00    | 906.30                          | 2435.36                                         | 1812.6                   | 238869                         |
|       |            |                                 | Productivities -->>                             |                          | 0.0076                         |

**Table A2-2 : Analysis of freeway segments on US 169**

No Ramp meters

| Freeway Sections               | VKTs for segments | VSTs for segments | Individual segment Productivities |
|--------------------------------|-------------------|-------------------|-----------------------------------|
| Valley View Rd-TH 62 EB        | 31743             | 1090966           | 0.0291                            |
| Th 62 EB - TH 62 WB            | 6562              | 353259            | 0.0186                            |
| TH 62 WB - Bren Road           | 11180             | 1156649           | 0.0097                            |
| Bren Road - 7th St EB          | 25816             | 1716304           | 0.0150                            |
| 7th St EB - Excelsior Blvd     | 18260             | 2161506           | 0.0084                            |
| Excelsior Blvd - TH 7          | 18681             | 1824628           | 0.0102                            |
| TH 7 - 36th Street             | 9012              | 718217            | 0.0125                            |
| 36th St - Minnetonka Blvd      | 13322             | 931758            | 0.0143                            |
| Minnetonka Blvd-Cedar Lake Rd  | 11640             | 814450            | 0.0143                            |
| Cedar Lake Rd-I 394 EB         | 41143             | 2675769           | 0.0154                            |
| I 394 EB - I 394 WB            | 5979              | 543483            | 0.0110                            |
| I 394 WB - Betty Crocker Drive | 5979              | 543483            | 0.0110                            |
| Betty Crocker Drive - TH 55 EB | 8573              | 690081            | 0.0124                            |
| TH 55 EB - TH 55 WB            | 7572              | 424683            | 0.0178                            |
| Th 55WB - Plymouth Avenue      | 19390             | 1170929           | 0.0166                            |
| Plymouth Av -Medicine Lake Rd  | 36538             | 1955474           | 0.0187                            |
|                                | 271388            | 18771640          | 0.0145                            |
|                                | Productivity-->   | 0.0145            |                                   |

# Ramp meters on

| Freeway Sections               | VSTs for segments | VKTs for segments | Individual<br>segment<br>Productivities | Changes in<br>consumer surplus<br>on individual<br>segments . |
|--------------------------------|-------------------|-------------------|-----------------------------------------|---------------------------------------------------------------|
| Valley View Rd-TH 62 EB        | 856555            | 27552             | 0.0322                                  | 465671                                                        |
| Th 62 EB - TH 62 WB            | 145951            | 5285              | 0.0362                                  | 314530                                                        |
| TH 62 WB - Bren Road           | 543705            | 15601             | 0.0287                                  | 1134105                                                       |
| Bren Road - 7th St EB          | 1248094           | 35036             | 0.0281                                  | 1112634                                                       |
| 7th St EB - Excelsior Blvd     | 935320            | 26049             | 0.0279                                  | 2097092                                                       |
| Excelsior Blvd - TH 7          | 923305            | 25936             | 0.0281                                  | 1611378                                                       |
| TH 7 - 36th Street             | 535857            | 12207             | 0.0228                                  | 432927                                                        |
| 36th St - Minnetonka Blvd      | 663453            | 17682             | 0.0267                                  | 570077                                                        |
| Minnetonka Blvd-Cedar Lake Rd  | 477248            | 15413             | 0.0323                                  | 597953                                                        |
| Cedar Lake Rd-I 394 EB         | 1758693           | 54268             | 0.0309                                  | 1775526                                                       |
| I 394 EB - I 394 WB            | 321730            | 7754              | 0.0241                                  | 376823                                                        |
| I 394 WB - Betty Crocker Drive | 321730            | 7754              | 0.0241                                  | 376823                                                        |
| Betty Crocker Drive - TH 55 EB | 490425            | 10842             | 0.0221                                  | 367928                                                        |
| TH 55 EB - TH 55 WB            | 301156            | 9483              | 0.0315                                  | 221088                                                        |
| Th 55WB - Plymouth Avenue      | 853308            | 24256             | 0.0284                                  | 571380                                                        |
| Plymouth Av -Medicine Lake Rd  | 1652405           | 44705             | 0.0271                                  | 688093                                                        |
|                                | 12028936          | 339822            |                                         | 12714028                                                      |
|                                | Productivity-->   | 0.0283            |                                         |                                                               |

**Table A2-3: Analysis of freeway segments on I 94:****No ramp meters**

| Segments              | VKTs for segments | VSTs for segments | Individual segment Productivities |
|-----------------------|-------------------|-------------------|-----------------------------------|
| Olson Hwy to Lyndale  | 79559             | 3462700           | 0.023                             |
| Lyndale to 5th Avenue | 49234             | 1395324           | 0.035                             |
| 5th Avenue to Cedar   | 27196             | 949427            | 0.029                             |
| Cedar to Riverside Av | 38178             | 1945085           | 0.020                             |
| Rivrside Av to Cretin | 116629            | 5058955           | 0.023                             |
| Cretin to Lexington   | 113157            | 3187741           | 0.035                             |
| Lexington to Dale     | 57726             | 3483817           | 0.017                             |
| Dale downstream       | 41345             | 1899700           | 0.022                             |
| System                | 523027            | 21382750          | 0.024                             |

**Ramp meters on**

| Segments              | VKTs for segments | VSTs for segments | Individual segment Productivities | Changes in consumer surplus in individual segments . |
|-----------------------|-------------------|-------------------|-----------------------------------|------------------------------------------------------|
| Olson Hwy to Lyndale  | 81589             | 3367837           | 0.024                             | 182757                                               |
| Lyndale to 5th Avenue | 49777             | 1386699           | 0.036                             | 25057                                                |
| 5th Avenue to Cedar   | 27607             | 801944            | 0.034                             | 162375                                               |
| Cedar to Riverside Av | 40613             | 1460700           | 0.028                             | 612910                                               |
| Rivrside Av to Cretin | 119028            | 3783893           | 0.031                             | 1390316                                              |
| Cretin to Lexington   | 116439            | 4486738           | 0.026                             | -1168157                                             |
| Lexington to Dale     | 60180             | 2544957           | 0.024                             | 1128572                                              |
| Dale downstream       | 44049             | 1945583           | 0.023                             | 124103                                               |
| System                | 539286            | 19778350          | 0.027                             | 2457934                                              |

**Table A2-4: Ramp Productivity on US 169.**

| Ramps               | With Ramp meters  |              |      |         |  |
|---------------------|-------------------|--------------|------|---------|--|
|                     | Flows             | Travel times | VKT  | VST     |  |
| Valley View Rd      | 2208              | 2943         | 382  | 166526  |  |
| TH 62 EB            | 2383              | 2194         | 429  | 123272  |  |
| Bren Road           | 1588              | 7363         | 400  | 273731  |  |
| Excelsior Blvd      | 1602              | 8583         | 519  | 318377  |  |
| TH 7                | 1417              | 4928         | 449  | 164301  |  |
| 36th Street         | 799               | 8067         | 207  | 151248  |  |
| Minnetonka Blvd     | 665               | 6591         | 110  | 106893  |  |
| Cedar Lake Road     | 551               | 6566         | 44   | 84973   |  |
| I 394 EB            | 1573              | 1546         | 340  | 60038   |  |
| I 394 WB            | 2051              | 7884         | 451  | 380243  |  |
| Betty Crocker Drive | 619               | 7409         | 45   | 115109  |  |
| TH 55 EB            | 635               | 9724         | 137  | 155883  |  |
| TH 55 WB            | 719               | 7776         | 223  | 137060  |  |
| Plymouth Avenue     | 908               | 4992         | 65   | 115165  |  |
| Medicine Lake Road  | 999               | 7514         | 194  | 177632  |  |
| System              |                   |              | 3994 | 2530451 |  |
|                     | Productivity -->> |              |      | 0.0016  |  |

| Ramps               | Without ramp meters |              |      |        | Changes in Consumer surplus |
|---------------------|---------------------|--------------|------|--------|-----------------------------|
|                     | Flows               | Travel times | VKT  | VST    |                             |
| Valley View Rd      | 1837                | 622          | 317  | 28572  | -117323                     |
| TH 62 EB            | 1837                | 648          | 331  | 29762  | -81548                      |
| Bren Road           | 1606                | 907          | 405  | 36428  | -257760                     |
| Excelsior Blvd      | 1780                | 1167         | 577  | 51910  | -313513                     |
| TH 7                | 1337                | 1141         | 424  | 38124  | -130378                     |
| 36th Street         | 719                 | 933          | 186  | 16775  | -135361                     |
| Minnetonka Blvd     | 756                 | 596          | 125  | 11269  | -106476                     |
| Cedar Lake Road     | 397                 | 285          | 31   | 2830   | -74426                      |
| I 394 EB            | 1142                | 778          | 247  | 22203  | -26060                      |
| I 394 WB            | 2008                | 792          | 442  | 39762  | -359847                     |
| Betty Crocker Drive | 829                 | 259          | 60   | 5372   | -129403                     |
| TH 55 EB            | 736                 | 778          | 159  | 14309  | -153325                     |
| TH 55 WB            | 741                 | 1115         | 229  | 20649  | -121569                     |
| Plymouth Avenue     | 1122                | 259          | 81   | 7271   | -120106                     |
| Medicine Lake Road  | 1041                | 700          | 202  | 18215  | -173760                     |
| System              |                     |              | 3816 | 343452 | -2,300,854                  |
|                     | Productivity -->>   |              |      | 0.0111 |                             |

**Table A2-5: Ramp Productivity on I-94**

| Ramps        | With Ramp meters  |              |      |         |
|--------------|-------------------|--------------|------|---------|
|              | Flows             | Travel times | VKT  | VST     |
| Olson Hwy    | 3873              | 2435         | 1812 | 238869  |
| Lyndale      | 1111              | 1902         | 240  | 53651   |
| 5th Avenue   | 1145              | 6280         | 305  | 176707  |
| Cedar avenue | 1409              | 4470         | 162  | 156635  |
| Riverside Av | 1718              | 3357         | 185  | 141685  |
| Cretin       | 2017              | 1655         | 305  | 78718   |
| Lexington    | 2597              | 1990         | 561  | 125760  |
| Dale         | 1653              | 1490         | 214  | 60817   |
| System       |                   |              | 3785 | 1032841 |
|              | Productivity -->> |              |      | 0.004   |

| Ramps        | Without ramp meters |              |      |        |               |
|--------------|---------------------|--------------|------|--------|---------------|
|              | Flows               | Travel times | VKT  | VST    | Changes in CS |
| Olson Hwy    | 3925                | 1685         | 1836 | 165338 | -73144        |
| Lyndale      | 1016                | 778          | 219  | 19753  | -29896        |
| 5th Avenue   | 1479                | 959          | 394  | 35464  | -174540       |
| Cedar avenue | 1497                | 415          | 172  | 15522  | -147318       |
| Riverside Av | 1767                | 389          | 190  | 17177  | -129305       |
| Cretin       | 1965                | 544          | 297  | 26742  | -55257        |
| Lexington    | 2376                | 778          | 513  | 46194  | -75357        |
| Dale         | 1508                | 467          | 195  | 17591  | -40418        |
| System       | 15533               |              |      | 343782 | -725,235      |
|              |                     |              |      | 0.011  |               |



### Appendix 3: Correlation Coefficients

Correlation coefficients between mobility and equity are shown in Table A3-1 for the both cases (ramp metering on and off). Statistically significant relationships between mobility and equity are found with respect to the following mobility measures: total travel time, travel time per mile, freeway travel time per mile, speed and freeway speed. Since the higher the Gini coefficient the lower the equity (perfect inequity = a Gini Coefficient of 1, perfect equity gives a Gini Coefficient = 0), Table 3 finds a positive relationship between mobility and equity when metering is on. A correlation coefficient of 1 indicates perfect correlation, a 0 indicates no relationship, and a -1 indicates perfect negative correlation. For instance, the higher the travel time per mile, the higher the Gini coefficient, or the lower the mobility, the more unfair the system is. In contrast, without metering the correlation is relatively weak. A hypothesis is that ramp meters can strengthen the relationship between mobility and equity.

Again within each case a positive correlation between mobility (most obviously, speed) and equity is found, seen in Table A3-1. On Route 169, an interesting result is the higher correlation in ramp on case than ramp off case. However, on I-94, this is not true. The conclusion that ramp meters can strengthen the relationship between mobility and equity cannot be drawn. More analyses on other freeway networks are necessary.

No statistically significant relationships between mobility and equity are found at entrance ramps on Route 169. Several reasons may account for the absence of a relationship:

- There may in fact be no relationship
- Ramp delays are calculated from two different types of data and this may skew delay calculation;
- Some upstream detectors may not provide correct counts.

Only 15 ramps were observed, this is a small sample, a larger sample may generate a relationship.

**Table A3-1 Correlation Coefficients Between Equity and Mobility**

| Mobility        | Metering             |        |                     |        | No Metering          |        |                     |        |
|-----------------|----------------------|--------|---------------------|--------|----------------------|--------|---------------------|--------|
|                 | Temporal Equity Gini |        | Spatial Equity Gini |        | Temporal Equity Gini |        | Spatial Equity Gini |        |
|                 | Route 169            | I-94   | Route 169           | I-94   | Route 169            | I-94   | Route 169           | I-94   |
| Travel Time     | -0.378               | -0.567 | -0.716              | -0.638 | 0.056                | -0.169 | 0.535               | 0.214  |
| 1/Speed         | 0.53                 | 0.767  | 0.814               | 0.846  | 0.323                | 0.84   | -0.158              | 0.927  |
| 1/Freeway Speed | 0.82                 | -0.03  | 0.937               | 0.726  | 0.468                | 0.84   | 0.253               | 0.927  |
| Speed           | -0.883               | -0.87  | -0.925              | -0.759 | -0.114               | -0.722 | 0.126               | -0.934 |
| Freeway Speed   | -0.749               | 0.02   | -0.96               | -0.732 | 0.02                 | -0.722 | -0.662              | -0.934 |
| Delay           | 0.127                | -0.568 | -0.348              | 0.244  | 0.005                | -0.266 | -0.025              | 0.109  |
| Delay/Distance  | -0.107               | 0.245  | -0.292              | 0.696  | 0.158                | 0.428  | -0.425              | 0.679  |

## APPENDIX 4: Headway Distributions

### Impacts of Different Headway Distribution on the Ramp Delay.

When we calculate the ramp delays, different headway assumption can give us different results. We are concerned with the impacts of different headway assumptions. In the Route 169 ramp delay calculation, two principal headway distributions are used, calculated, and the delay results are compared:

**Exponential Distribution:** Time intervals between two consecutive vehicle arrivals at ramps are sampled from an exponential distribution (within each five minute period). The mean input flow (in vehicles/second) is  $\lambda$ , and the mean time headway is calculated as  $1/\lambda$  seconds.

**Uniform Distribution:** Time intervals between two consecutive vehicle arrivals at ramps are always constant (within each five minute period). The headway is taken as  $1/q$  seconds, being  $q$  the mean input flow (in vehicles/second).

The results of total ramp delay, number of delayed vehicles and Gini coefficient for all the different distribution are attached in the file Arrival Distribution Comparison.xls. We found the following rules:

#### **(1) Delay Uniform $\leq$ Delay Exponential**

Although the differences of delays among the three distributions are not very large, this relationship holds for all the 15 ramps.

#### **(2) Delayed Vehicles Uniform $\leq$ Delayed Vehicles Exponential**

In the exponential assumption, all the headways are sampled by assigning an average of several random processes (depends on the simulation times), so almost all the vehicles will be delayed. The number of delayed vehicles in the uniform distribution assumption is less.

#### **Gini Constant $\approx$ Gini Exponential**

From the results of the 15 ramps, neither the Gini coefficient with a uniform delay nor the Gini coefficient exponential is always larger than the other

**Table A4-1 Ramp Delays under Different Headway Assumptions**

| Ramps              | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>Exponential</b> |       |       |       |       |       |       |       |       |       |
| Total Delay(h)     | 25.18 | 9.38  | 43.87 | 42.06 | 37.96 | 15.76 | 23.55 | 12.41 | 22.56 |
| Delayed veh        | 3873  | 1111  | 1143  | 1407  | 1718  | 2017  | 2597  | 1653  | 2186  |
| Gini               | .5098 | .4512 | .3299 | .3241 | .5681 | .5467 | .4494 | .4825 | .5456 |
| <b>Uniform</b>     |       |       |       |       |       |       |       |       |       |
| Total Delay        | 20.06 | 6.24  | 43.20 | 41.32 | 35.59 | 11.59 | 20.33 | 8.59  | 18.91 |
| Delayed veh        | 2709  | 618   | 1084  | 1403  | 1326  | 911   | 1910  | 871   | 1383  |
| Gini               | .5736 | .5494 | .3089 | .343  | .5367 | .5337 | .4499 | .5472 | .5439 |

## **Appendix 5: Accessibility Methodology**

The average speed for each freeway segment is obtained by taking averages of the average space mean speed. This means that first the average speed of each detector station is obtained for the peak hour period. Then depending upon OD pairs, the average freeway speed is obtained by summing up the upstream and down stream detectors station. This will give us the average space mean speed for that freeway section. As an example, the average speed between Cretin Avenue and Dale Street is shown in Table A5-1. This average speed is then used to approximate the travel time between the exit ramps.

Table A5-2 gives the new OD pairs considered for calculation of accessibility. These replace the OD pairs used in the main document because either there is no exit ramp at that location or they are far away from the ramps under consideration. These new pairs will give us a more accurate estimate of travel time.

**Table A5-1: Calculation of average speeds  
I-94 Ramp On Case**

| Highway        | Start Station | Detectors | Speed  | Speed from Formula 1 | Volume | Weighted Space mean speed | Avg. Speed |
|----------------|---------------|-----------|--------|----------------------|--------|---------------------------|------------|
|                |               |           |        |                      |        |                           |            |
| Lexington ave. | 489           | 2638      | 60.970 | 72.055               | 3472   |                           |            |
|                |               | 2639      | 59.610 | 70.448               | 4994   |                           |            |
|                |               | 2640      | 54.732 | 64.683               | 5663   |                           |            |
|                |               | 2641      | 70.567 | 83.397               | 6468   |                           |            |
|                |               |           |        |                      | 20597  | 73.200                    | 63.560     |
|                |               |           |        |                      |        |                           |            |
| Dale St.       | 490           | 2643      | 57.802 | 68.311               | 4035   |                           |            |
|                |               | 2644      | 56.499 | 66.771               | 4574   |                           |            |
|                |               | 2645      | 59.843 | 70.724               | 5794   |                           |            |
|                |               | 2646      | 63.472 | 75.012               | 6798   |                           |            |
|                |               |           |        |                      | 21201  | 70.787                    | 71.994     |
|                |               |           |        |                      |        |                           |            |
| Marion St.     | 491           | 2648      | 59.578 | 70.410               | 5466   |                           |            |
|                |               | 2649      | 56.137 | 66.343               | 6366   |                           |            |
|                |               | 2650      | 54.576 | 64.499               | 7536   |                           |            |
|                |               |           |        |                      | 19368  | 66.774                    | 68.780     |

**Table A5-2. New OD pairs**

| OLD pairs                 | New pairs          |
|---------------------------|--------------------|
| Olson Hwy - Lyndale       | Olson hwy - I-394  |
| Olson hwy - 5th Ave.      | Olson hwy - I35W   |
| Olson hwy - cedar         | Olson hwy - TH55   |
|                           | Olson hwy - 25th   |
| Olson hwy - riverside     | Ave.               |
| Olson hwy - cretin        | Olson hwy - cretin |
|                           | Olson hwy -        |
| Olson hwy - Lexington     | Lexington          |
| Olson hwy - dale          | Olson hwy - dale   |
| Olson hwy - Marion        | Olson hwy - Marion |
| 5th Ave. - cedar Ave.     | I35W - TH 55       |
| 5th Ave. - riverside Ave. | I35W - 25th Ave    |
| 5th Ave. - cretin Ave.    | I35W - cretin      |
| 5th Ave. - Lexington      |                    |
| Ave.                      | I35W - Lexington   |
| 5th Ave. - dale           | I35W - dale        |
| 5th Ave. - Marion         | I35W - Marion      |
| Cedar Ave. - riverside    |                    |
| Ave.                      | TH55- 25th Ave.    |
| Cedar Ave. - cretin Ave.  | TH55- cretin       |
| Cedar Ave. - Lexington    | TH55- Lexington    |
| Cedar Ave. - dale         | TH55- dale         |
| Cedar Ave. - Marion       | TH55- Marion       |
| Riverside - cretin        | 25th - cretin      |
| Riverside - Lexington     | 25th - Lexington   |
| Riverside - dale          | 25th - dale        |
| Riverside - Marion        | 25th - Marion      |
| Cretin - Lexington        | Cretin - Lexington |
| Cretin - dale             | Cretin - dale      |
| Cretin - Marion           | Cretin - Marion    |
| Lexington - dale          | Lexington - dale   |
| Lexington - Marion        | Lexington - Marion |
| Dale - Marion             | Dale – Marion      |

## APPENDIX 6: SOURCES OF POSSIBLE ERROR

The queue lengths for a few of the ramps (3A3, 3C1 and 3C3) were found to be negative over short time periods. This can happen only if the number of arrivals measured during a particular interval of time is less than the number of departures plus the previous queue length. There are several possibilities:

- The detectors were not working. But if it were the cause, then they would not be giving any count at all, rather than giving some low count (possibly ..) as now. It is unlikely the detectors would be in a state in between working and not-working. And they would not be also behaving intermittently as working fine at some times and not working at other times....(This might be a remote possibility though).
- Cars enter the ramp somewhere in between the arrival and departure detectors. If the ramp detectors are placed badly, this may occur.
- Cars arriving on the ramp could sidetrack the loop detectors, and as a result the detectors would not pick up the signal. If the detectors were properly placed, the cars would not really sidetrack the detectors (considering that the detectors are placed in the middle of the road) unless they were banking at a high speed, which is rare, or they would not be able to take the center of the road as it was already occupied by other cars forming a queue over the middle of the road, where the detector is located. But detectors are able to pick up signals from passing cars even if the cars were not passing directly over the detector. A 6 ft by 6 ft detector is able to track any changes in the magnetic field up to 3 ft on either side. Normally the on ramps are hardly more than 18 ft wide and cars are certainly greater than 3 ft wide. So that leaves only 3 ft for the car to veer through on each side of the road without being detected, which they can't possibly do (as their width is more than 3 ft,) So their influence on the detector would be felt some way or the other. However if the number of loops in the detector are less than 3 or 4, the detector however might be weak and unable to detect cars passing far from it. Again this is a singular possibility only. If the roads are covered by a wider loop than 6 ft by 6 ft, then not detecting the cars would be more unlikely.

This leads to another possibility of errors. While calculating we had neglected the arrival rates of the HOVs as they were very few, and they would not be forming a queue. But they may have passed over the departure loop detectors (depending on location) which would then give their counts too, which we had neglected while considering the arrivals. This could have led to the departure rates being shown more than the arrival rates, which can't be possible in reality. So there might be an option for computing the queue lengths considering the passage of the HOVs too and checking out the real working condition of the detectors. Again this depends on the physical configuration of the ramps.



## APPENDIX 7: NON-LINEAR VALUE OF TIME

The literature on value of travel time is vast, but the part of that literature that focuses on differentiating travel time by the quality of the trip is almost non-existent. Some mode choice literature values travel times by transit legs separately (access, wait, and in-vehicle transit time) (e.g. Ben-Akiva and Lerman 1985), but no similar study exists on the highway side.

Hensher (ITS-WP-00-9) considers that the empirical valuation of travel time savings is a derivative of the ratio of parameter estimates in a discrete choice model. The most common formulation (multinomial logit) imposes strong restrictions on the profile of the unobserved influences on choice as represented by the random component of a preference function. In this paper he contrasts the values of travel time savings derived from five discrete choice models – multinomial logit, heteroskedastic extreme value, covariance heterogeneity logit, mixed (or random parameter) logit and multinomial probit. The empirical setting is urban car commuting and non-and non-commuting in six locations in New Zealand. Similarly, The Preliminary Finnish Value-of-Time Study for Long Distance Travel explored the theory behind the assessment of and to work out models for the estimation of the value of travel time. Data of the survey of the Finnish high speed railway alternatives north-east of Helsinki are used in the modeling. The value of time is derived from a mode choice model. The attributes of modes are travel cost and time, and departures per day. Taste variations in values of time are examined by segmenting the observations by trip purpose, mode, and personal gross income. The ratio of value of time and income is not constant.

Bone (1987) reviews the practice of ascribing a dollar value to travel time savings arising from transport investment projects. The concept of time value is first discussed, drawing a distinction between its application in predicting travel demand and in economic evaluation. The paper suggests that the present uniform treatment of time savings does not allow level of service measures of the transport system to enter the economic evaluation process explicitly, although value of time implicitly includes some measure of the utility of travel. He argues that it would be desirable to separate the value of the travel experience from the opportunity value of the time saved although this may be difficult to achieve.

Hensher treats the topic (ITS-WP-00-17) of the behavioral value of travel time savings (VTTS). Numerous studies have been conducted to shed light on how much an individual or firm is willing to pay to save a unit of time, the majority of empirical studies have used essentially the same data metric and model estimation procedure. The most popular approaches have used either revealed preference (RP) or stated preference (SP) data and a multinomial logit (MNL) choice model to identify marginal rates of substitution between travel time and price of a trip. This paper highlights the potential gains from a more carefully considered structure of the unobserved effects conditioning the form of a discrete choice model, and hence the possible misinterpretation from the simpler Multinomial Logit (MNL) specification. It demonstrates the implication for

VTTS of serial correlation between the SP treatments, covariance amongst alternatives, the presence of individual specific (random) effects or heterogeneity, and differential variance of the unobserved effects.

David Hensher and Truong P Truong ( ), note that travel time can't be saved, it can only be transferred from one activity to another. This point is reinforced in Levinson and Kanchi (2000), who study the changes in travel and activity times resulting from additional infrastructure.

Algers, Bergstrom, Dahlberg and Dillen (Mixed logit estimation of value of travel time, 1998) argue that the value of time is calculated as a trade off between in-vehicle time coefficient and the cost coefficient. A common segmentation is the trip purpose.. work trips or business trips, Sometimes trip lengths and income levels are also used to differentiate value of time in trips. Here the authors apply mixed logit specifications to allow parameters to vary in the population when estimating the value of time for long distance travel. The main conclusion that the authors draw is that the value of time is highly sensitive to the way the model is specified. It is found that the value of time is significantly lower when the coefficients are assumed to be normally distributed in the population in contrast to the traditional case where they are treated as fixed. Also when the authors allowed the parameters to vary log normally in the population, it was found that the models did not converge unless the dispersion parameter of the log-normal distribution was constrained.

Johansson ( Paper III, of "Four papers on the contingent valuation and stated preference methods,1999) uses stated preference data for the estimation of time and safety for work trips. In standard and random parameters logit models, the respondent's modal choice is related to the perceived time, cost, safety and the environment friendliness of the available alternatives. These parameters are shown to have a significant effect on the choice of work trip travel mode. From the estimated models, the attribute values of time and safety are assessed.

Kawamura ( Urban Planning and Policy program, UIC, TRB 79<sup>th</sup> meet, Washington DC) estimated the value of time for trucks from the stated preference data collected in California. Truckers were asked about a choice between an existing free road versus a toll facility for different combinations of travel time and cost. Estimation was based on the point of diversion at which the switch of facility occurred in the stated preference questions, and also using a modified logit model in which the coefficients to be estimated were assumed to be distributed lognormally across the population. The former approach revealed that the value of time can be well replicated with a lognormal distribution. of time. The author draws lines of difference between the three methods hitherto used to measure the value of travel time saved by the truck operators: 1) the **cost savings method**, which is based on the cost savings to operators per unit of time, 2) the **revenue (net profit) method**, which estimates the net increase in profit resulting from the reduction in travel time (1), and 3) the **willingness to pay method**, which measures the "perceived" value of time from observed or stated choices under trade-off situations involving time and money.

Casey, Vukina and Danielson (1995) test two alternative specifications for the opportunity cost of time in travel cost models. The standard travel cost survey design is enriched to include a contingent valuation type question about peoples' willingness to accept compensation to forgo a precisely defined recreational experience. It is hypothesized that individually revealed value of time more appropriately reflects the opportunity costs of time associated with a particular aspect of recreation than the wage rate which measures the trade-off between work and leisure, generally. Their results indicate a better overall fit for the models with the elicited value of individual consumer's time than for the models with the more traditional hourly earnings (wage rates). The importance of the correct measurement of the opportunity cost time is illustrated by showing that estimated consumer surpluses based on two different value of time measurements differ significantly.

Bone (1987) reviews the practice of ascribing a dollar value to travel time savings arising from transport investment projects. The concept of time value is first discussed, drawing a distinction between its application in predicting travel demand and in economic evaluation. Present New Zealand practice is reviewed and its relationship to practice overseas. The general findings of recent British studies are noted and the initial stages of an investigation sponsored by the New Zealand urban transport council are described. The paper suggests that our present uniform treatment of time savings does not allow level of service measures of the transport system to enter the economic evaluation process explicitly, although value of time implicitly includes some measure of the utility of travel. He suggests separation of the value of the travel experience from the opportunity value of the time saved although this may be difficult to achieve. Some doubts are expressed as to the validity of suppositions embodied in travel choice models used to derive travel time values. It is argued that the acceptance or rejection of an equity (or average) opportunity value of time savings is an important matter of policy which should be decided before any move to ascribe differing values of non-work time according to socio-economic characteristics of the travelling public

Knudson (Oregon Department of transportation) conducted studies for 2 years (1997, 98) in order to report and document estimated values of travel-time for vehicles in Oregon. Studies within ODOT have used a number of different estimates for the value of travel-time. Costs associated with travel are considered to be of 2 primary types...Vehicle operating costs and the value of travel time. The study basically focuses on the costs associated with travel time. Travel times behaving in nonlinear fashion are estimated depending upon purpose of trip, selection and availability of relevant data, methodology and underlying assumptions when building the estimate and the selection and availability of detailed data. Costs associated with travel-time include employers' cost of employees traveling on-the-job, depreciation of inventory, and spoilage cost of goods being transported. For other drivers, travel-time costs include the opportunity cost of their time spent driving – time that could be spent doing something else. The value of one hour of travel-time is estimated for six vehicle categories using national data. Further estimates are made using Oregon data for three more general vehicle categories.

Cohen and Southworth (1999) try to estimate the dollar value of travel time due to unanticipated freeway incidents. Apart from providing a simple model for estimating the mean and variance of time lost due to incidents on freeways also reviews methods for assigning a monetary value to the variability that such incidents induce to daily travel. The authors argue that avoiding the time spent in travelling is a non-productive activity against which there is an opportunity cost. A common approach to placing a cost to the extra time spent in travel is to assess the value of travel time in terms of hours lost multiplied by some fraction of gross hourly wage. The authors assign a method to measuring the travel time variability from an incident analysis procedure.

Small, Noland, Chu and Lewis (1999) try to develop methodologies for measuring the effects of congestion on the values highway users place on travel-time savings and predictability. The methodologies were used to generate values for factors for different degrees of congestion. The study also defines an approach for incorporating these factors in highway user-cost estimates.

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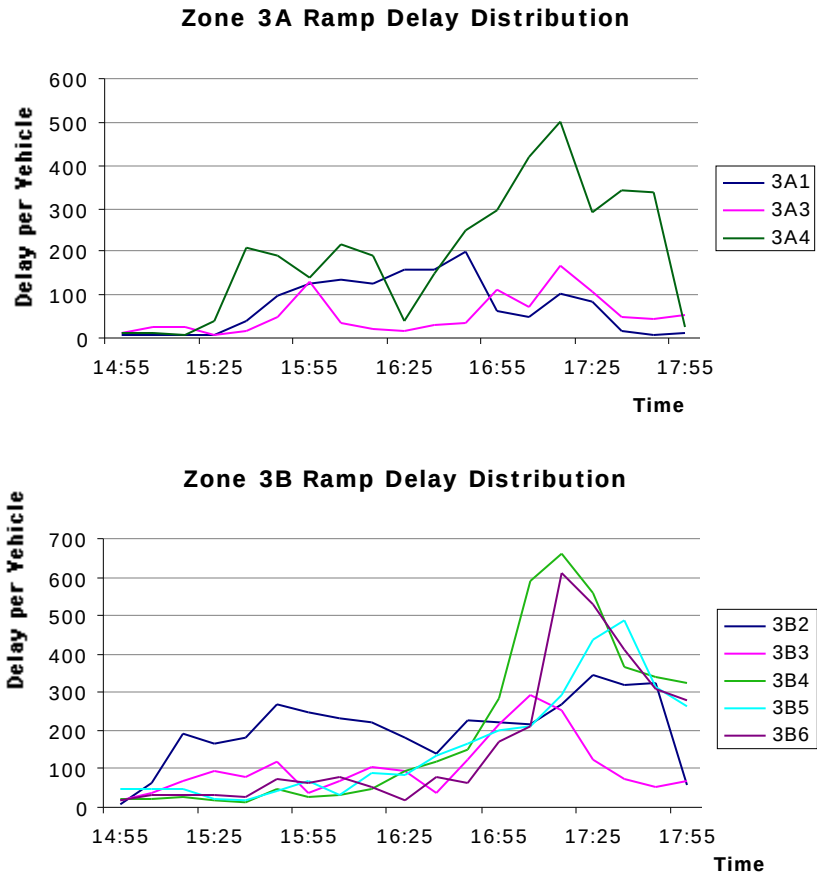
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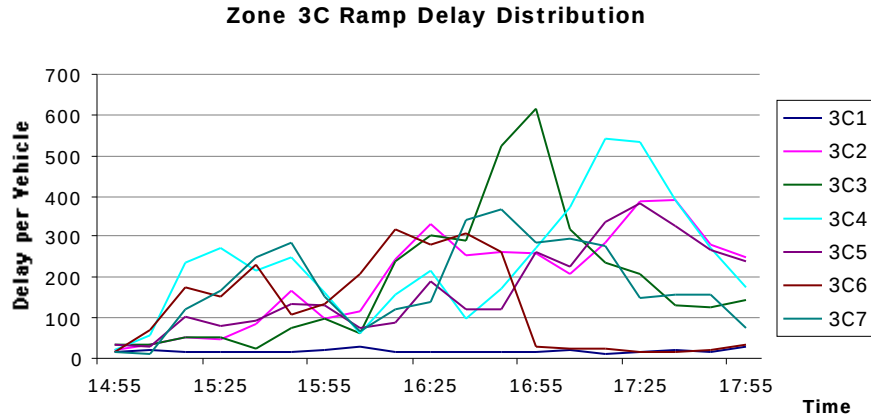
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## Appendix 8: HW 169 Zone 3A, 3B & 3C PM Peak Ramp Delay Distribution

Ramp delay distributions in freeway zone 3A, 3B and 3C on HW 169 are shown graphically in figure 1.





It seems that ramp delays in PM peak hours (14:55 to 18:10pm) were not distributed evenly within each zone. Particularly, as travel volumes become very high from 16:55 to 17:55pm, one can find one or two on-ramps in each zone which have much longer delays than others.

Two things can be done to improve it:

1. Relocate existing ramp arrival detectors to obtain more accurate traffic counts and install arrival detectors on ramps where it hasn't been installed, because it's almost impossible to distribute delays evenly without accurate real time demand volume data.
2. A better ramp coordination algorithm.



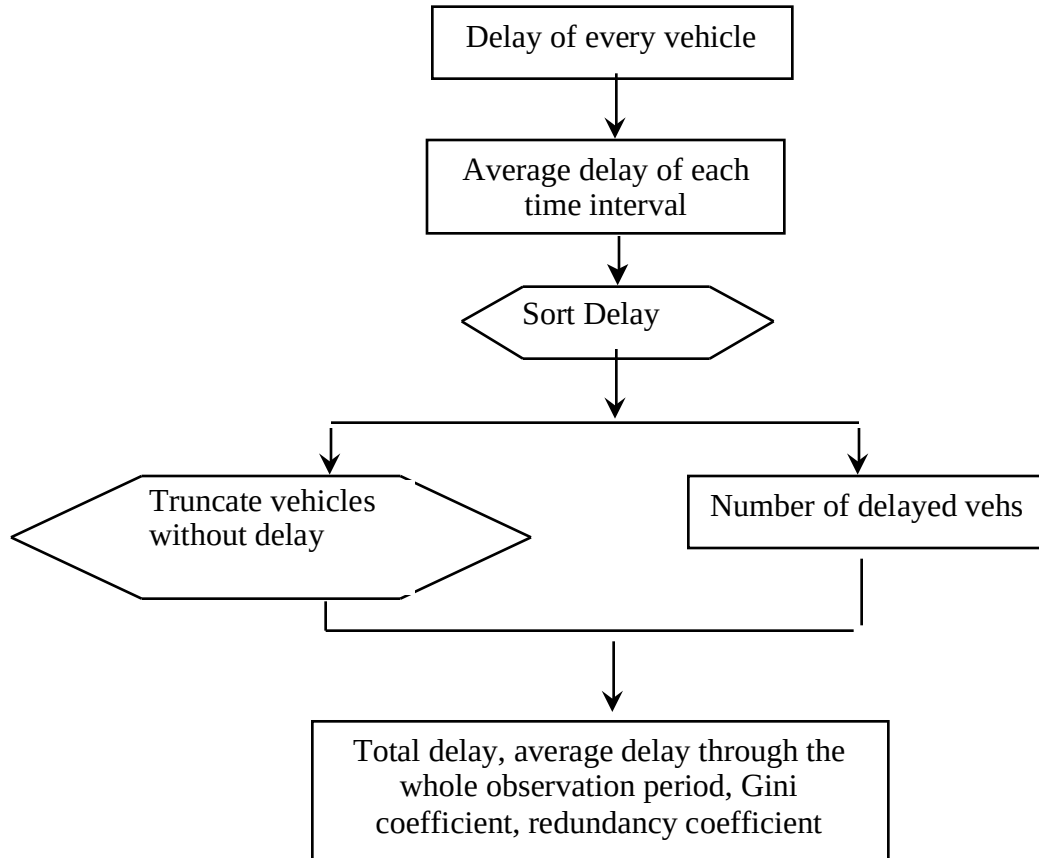
## Appendix 9: Vehicle Indexing Method

To illustrate the vehicle indexing method, consider the tables below. In Table A9-1, column B shows the time at which a particular car arrives. For example, car 1 finally arrives at the 9th second from the start of metering we can see that the figure '1' appears finally on the corresponding cell in the column C. Similarly the 1st car finally leaves in the 12th second from the start of metering. The Right Side of Table 1 illustrates the vehicle indexing method. Column P shows the index number of the individual cars and the same final arrival and departure times for each of the cars is shown in the columns Q and R separately. Finally the column S shows the delay of the corresponding individual cars. Vehicle 3 is highlighted with boxes for tracking.

**Table A9-1: Spreadsheet of Vehicle Indexing Method**

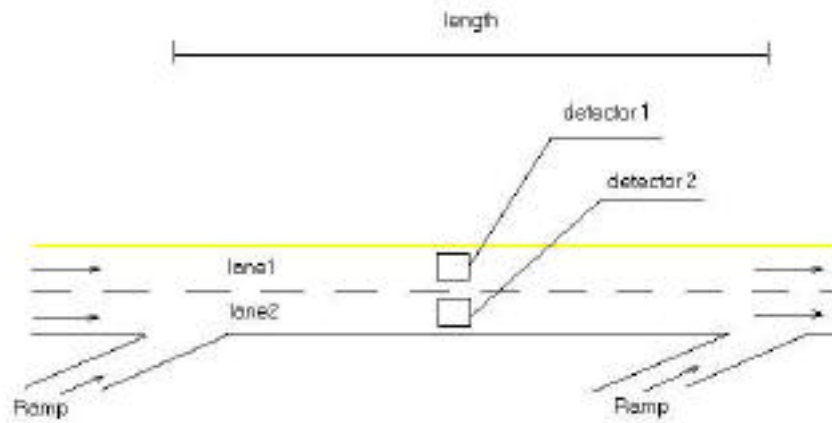
| A<br>1 Second<br>Intervals<br>From The<br>Time Queue<br>Forms | B<br>Time At<br>Which The<br>Individual<br>Cars Arrive | C<br>Time At<br>Which The<br>Individual<br>Cars Depart | P<br>Serial Numbers<br>Of Individual<br>Cars. | Q<br>The Time In<br>Seconds At<br>Which Each<br>Car Finally<br>Departs. | R<br>The Time In<br>Seconds At<br>Which Each<br>Car Finally<br>Arrives. | S<br>Delays Of<br>Each<br>Individual Car. |
|---------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------|
| 1                                                             | #n/a                                                   | #n/a                                                   | 1                                             | 12                                                                      | 9                                                                       | 3                                         |
| 2                                                             | #n/a                                                   | #n/a                                                   | 2                                             | 18                                                                      | 14                                                                      | 4                                         |
| 3                                                             | #n/a                                                   | #n/a                                                   | 3                                             | 24                                                                      | 19                                                                      | 5                                         |
| 4                                                             | #n/a                                                   | #n/a                                                   | 4                                             | 31                                                                      | 24                                                                      | 7                                         |
| 5                                                             | 1                                                      | #n/a                                                   | 5                                             | 37                                                                      | 29                                                                      | 8                                         |
| 6                                                             | 1                                                      | #n/a                                                   | 6                                             | 43                                                                      | 33                                                                      | 10                                        |
| 7                                                             | 1                                                      | 1                                                      | 7                                             | 49                                                                      | 38                                                                      | 11                                        |
| 8                                                             | 1                                                      | 1                                                      | 8                                             | 56                                                                      | 43                                                                      | 13                                        |
| 9                                                             | 1                                                      | 1                                                      | 9                                             | 62                                                                      | 48                                                                      | 14                                        |
| 10                                                            | 2                                                      | 1                                                      | 10                                            | 68                                                                      | 53                                                                      | 15                                        |
| 11                                                            | 2                                                      | 1                                                      | 11                                            | 74                                                                      | 58                                                                      | 16                                        |
| 12                                                            | 2                                                      | 1                                                      | 12                                            | 81                                                                      | 62                                                                      | 19                                        |
| 13                                                            | 2                                                      | 2                                                      | 13                                            | 87                                                                      | 67                                                                      | 20                                        |
| 14                                                            | 2                                                      | 2                                                      | 14                                            | 93                                                                      | 72                                                                      | 21                                        |
| 15                                                            | 3                                                      | 2                                                      | 15                                            | 99                                                                      | 77                                                                      | 22                                        |
| 16                                                            | 3                                                      | 2                                                      | 16                                            | 106                                                                     | 82                                                                      | 24                                        |
| 17                                                            | 3                                                      | 2                                                      | 17                                            | 112                                                                     | 87                                                                      | 25                                        |
| 18                                                            | 3                                                      | 2                                                      | 18                                            | 118                                                                     | 91                                                                      | 27                                        |
| 19                                                            | 3                                                      | 3                                                      | 19                                            | 124                                                                     | 96                                                                      | 28                                        |
| 20                                                            | 4                                                      | 3                                                      | 20                                            | 131                                                                     | 101                                                                     | 30                                        |
| 21                                                            | 4                                                      | 3                                                      | 21                                            | 137                                                                     | 106                                                                     | 31                                        |
| 22                                                            | 4                                                      | 3                                                      | 22                                            | 143                                                                     | 111                                                                     | 32                                        |
| 23                                                            | 4                                                      | 3                                                      | 23                                            | 149                                                                     | 116                                                                     | 33                                        |
| 24                                                            | 4                                                      | 3                                                      | 24                                            | 156                                                                     | 120                                                                     | 36                                        |
| 25                                                            | 5                                                      | 4                                                      | 25                                            | 162                                                                     | 125                                                                     | 37                                        |
| 26                                                            | 5                                                      | 4                                                      | 26                                            | 168                                                                     | 130                                                                     | 38                                        |
| 27                                                            | 5                                                      | 4                                                      | 27                                            | 174                                                                     | 135                                                                     | 39                                        |
| 28                                                            | 5                                                      | 4                                                      | 28                                            | 181                                                                     | 140                                                                     | 41                                        |
| 29                                                            | 5                                                      | 4                                                      | 29                                            | 187                                                                     | 145                                                                     | 42                                        |
|                                                               |                                                        |                                                        | 30                                            | 193                                                                     | 149                                                                     | 44                                        |

## Appendix 10: Miscellaneous Figures

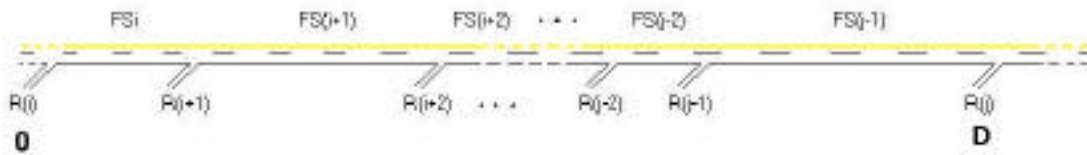


**Figure A10-1: Flowchart of Program**

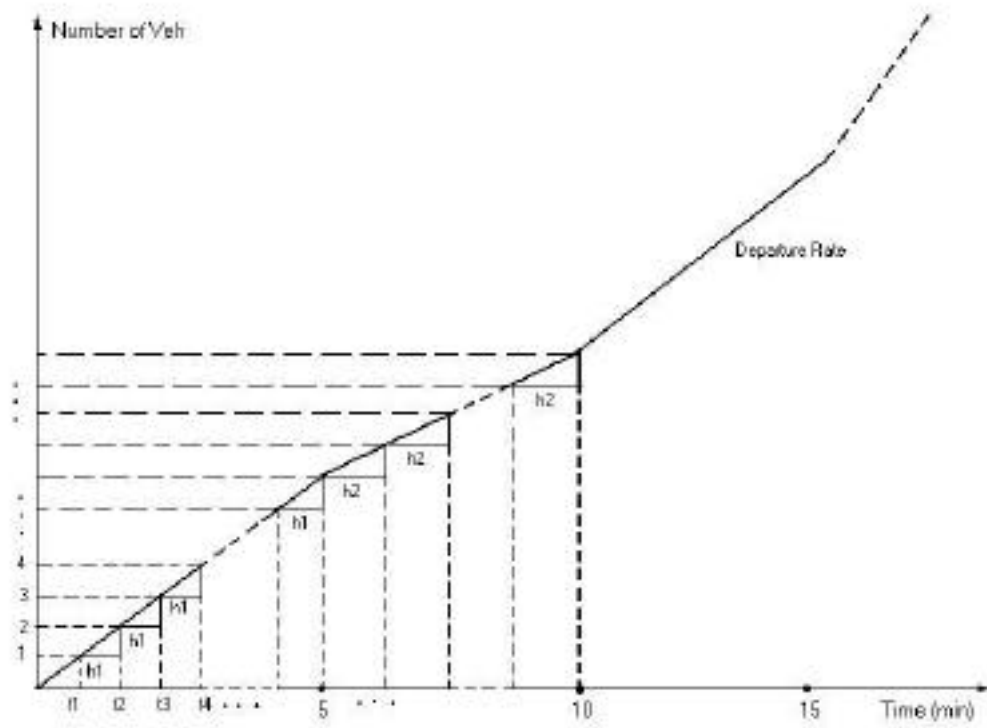




**Figure A10-3: Illustration of Freeway Detectors**



**Figure A10-4: Building Trip Times**



**Figure A10-6: Detail of Departure Rate**